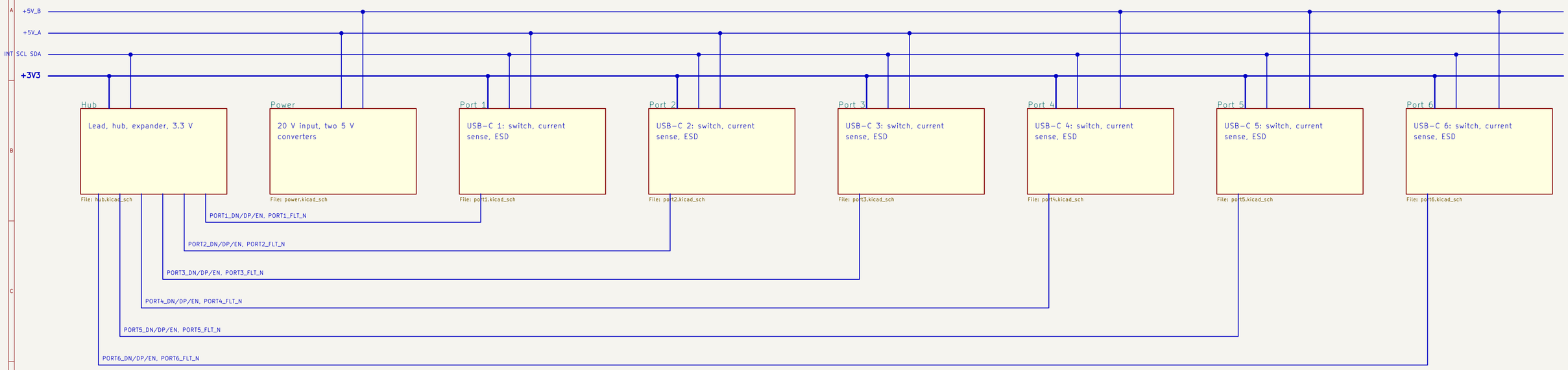


DATA ×6 v1.0 — USB2517I hub (7 ports, 6 used), six USB-C each with a TPS2553-1, an INA226 and ESD, a TCA9535 for their EN and FAULT_N, an AP2114H 3.3 V; two LMR51625 make the ports' 5 V from VIN (20 V, PH 3); the hub runs on the lead's switched 5 V (J1.8), as the v2's. No pin header: the porte-revues bays' boards take no Dupont lead (the DATA ×4 v2 keeps the logic). One face assembled.
Lead pinout here: 1 INT · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 +5V — the DATA ×4 v2's lead.
I²C: TCA9535 0x20, INA226 0x41 / 0x43 / 0x44 / 0x45 / 0x47 / 0x4C; the node's 0x40.
Rails: VIN 20 V (13.5-28 V; on above 13.5 V), ≤ 0.85 A at 20 V · +5V_A / +5V_B 5.24 V, ≤ 1.86 A each (ports 1-3 / 4-6 at the 0.62 A limit) · +5V_L 4.75-5.24 V (the node's switched 5 V over the lead), ≤ 0.46 A · PORTn_VBUS 5.1-5.24 V, 0.50-0.62 A limit · +3V3 3.3 V (AP2114H), ≤ 0.5 A (the hub 0.42 A at most) · HUB_1V8 / HUB_1V8PLL 1.8 V (the hub's own regulators).

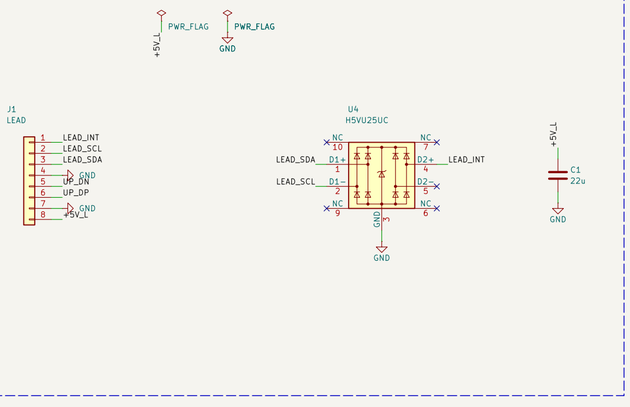
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)
hw-testing-rack

Sheet: /
File: data-port-x6.kicad_sch

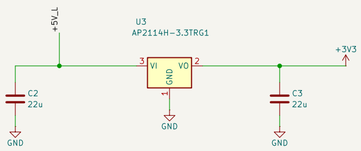
Title: DATA ×6 — Overview

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/9

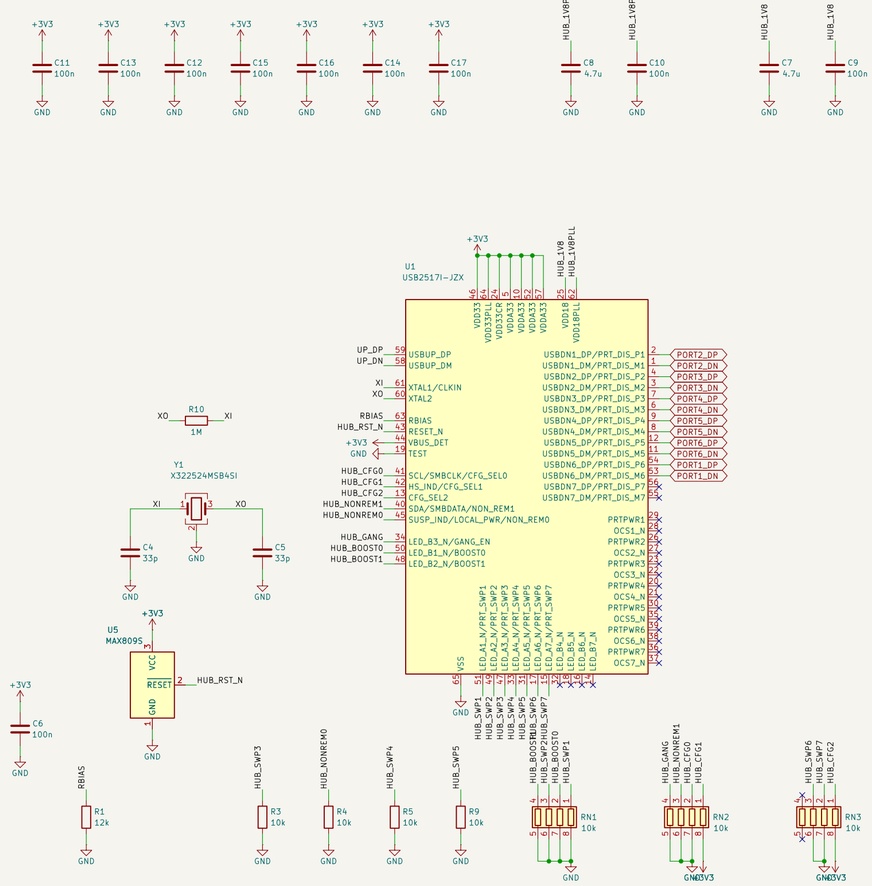
Lead from the node's J7 (PH 8, rear face): I²C and INT through an H5VU25UC; pin 8 the hub's 5 V



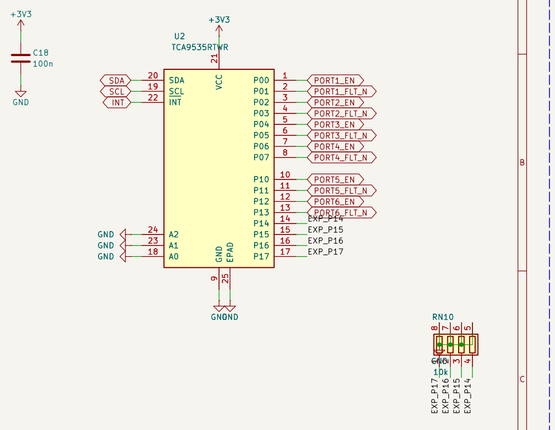
3.3 V for the hub, the expander and the INA226s (AP2114H, SOT-223 on copper)



USB2517I hub, CFG_SEL 110 (internal default configuration, straps disabled), 24 MHz crystal



TCA9535 expander (0x20): each port's EN and FAULT_N; its spare pins pulled down

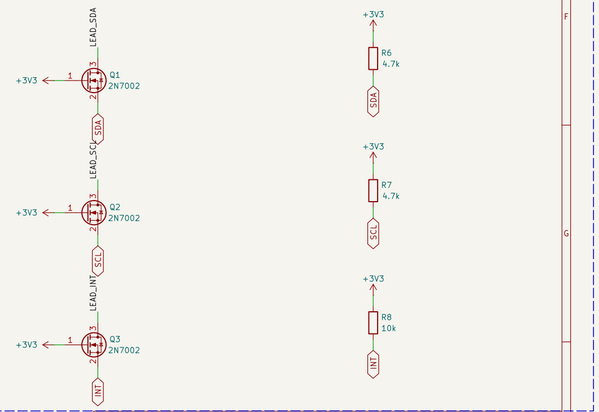


Hub reset: U5 (MAX809S, 2.93 V, ≥ 150 ms, push-pull) on RESET_N (errata DS80000628A, Module 2).
Hub: CFG_SEL[2:0] = 110 (internal default configuration, strap options disabled: DS00001598C Table 5-2).
as the DATA ×4 v2's: ports 1-6 used, port 7 enabled with nothing on it (cosmetic).
Hub ports: 1-5 = USB-C 2-6 (busid 1-1.1 … 1-1.5), 6 = USB-C 1 (1-1.6).
Expander: TCA9535 U2 0x20. Port n's EN on P0/P1(2n − 2). FAULT_N on the next bit (ports 1-4 on P00-P07,
5-6 on P10-P13). Inputs at power-up; the EN pull-downs keep every port off until the node writes; the
spare pins P14-P17 set as outputs, low, by the firmware.
INT (J1.1 → the node's IO4): the expander's INT and the six INA226 ALERTs, open drain, wired-OR, through Q3.
I²C: TCA9535 0x20, INA226 0x41 / 0x43 / 0x44 / 0x45 / 0x47 / 0x4C; the node's INA226 0x40.
J1 pin 8 (the node's switched +5V, ≤ 1.34 A) feeds the hub's 3.3 V (+5V_L: U3); the ports take theirs
from VIN (sheet Power). With +5V_L off, the expander's pins float and the EN pull-downs keep the ports off.
J1 (through-hole) stands on the rear face: soldered by hand (the PCBA places the front face only).

Test points (rear face, bare pads, not in the BOM): +3V3, the hub's reset and 1.8 V, the lead's I²C and INT, GND



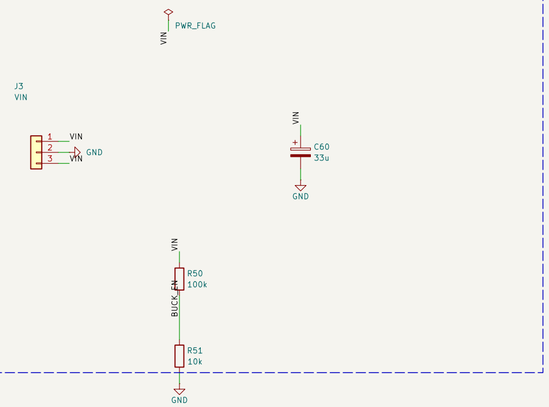
I²C isolation: open with the board's 3.3 V off (the node's pull-ups can't feed it)



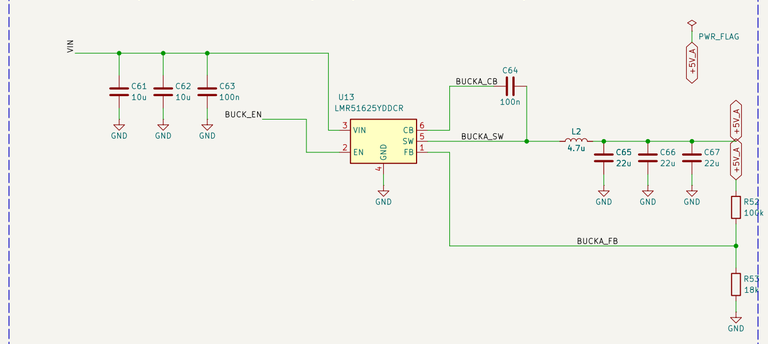
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)

hw-testing-rack		
Sheet: /Hub/		
File: hub.kicad_sch		
Title: DATA ×6 — Lead, hub, expander, 3.3 V		
Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 2/9

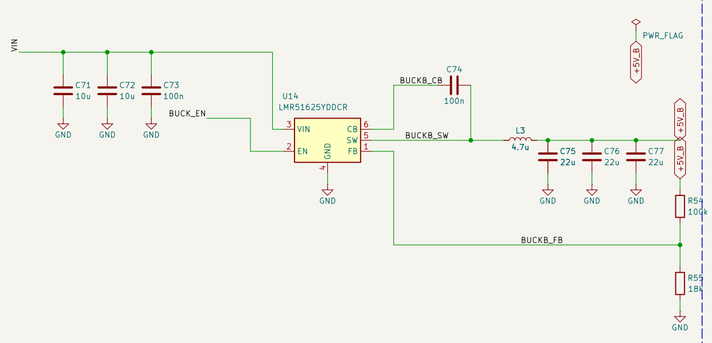
VIN: the rack's 20 V from the node (after its reverse protection), a PH 3 VIN-GND-VIN on the rear face



+5V_A: LMR51625, synchronous, 1.1 MHz, 2.5 A (the bay node's block; TI SLUSF64B)

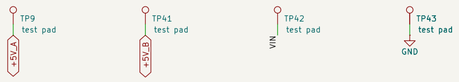


+5V_B: LMR51625, synchronous, 1.1 MHz, 2.5 A (the bay node's block; TI SLUSF64B)



Two LMR51625 as the bay node's (TI SLUSF64B table 8-1: 4.7 µH FXL0630, 2 × 10 µF + 100 nF in, 3 × 22 µF out).
+5V_A: ports 1-3; +5V_B: ports 4-6 (the logic takes +5V_L, J1.B). Never in parallel: the power plane's +5V split left / right. Each 1.5 A at 0.5 A a port, 1.86 A at the TPS2553s' 0.62 A limit (≈ 1 W lost each).
VIN on J3 (PH 3: 1 VIN, 2 GND, 3 VIN): either pre-crimped lead direction fits.
FB 100k/18k: 5.24 V, as the node's. EN 100k/10k: on above 13.5 V, off below 10.8 V.
VIN from the node's VRACK after its reverse-protection P-FET (node rev. B): no second protection here.
≈ 0.95 A at 20 V, 0.7 A at 28 V: a PH contact carries 2 A.

Test points (rear face): +5V_A, +5V_B, VIN, GND



Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)

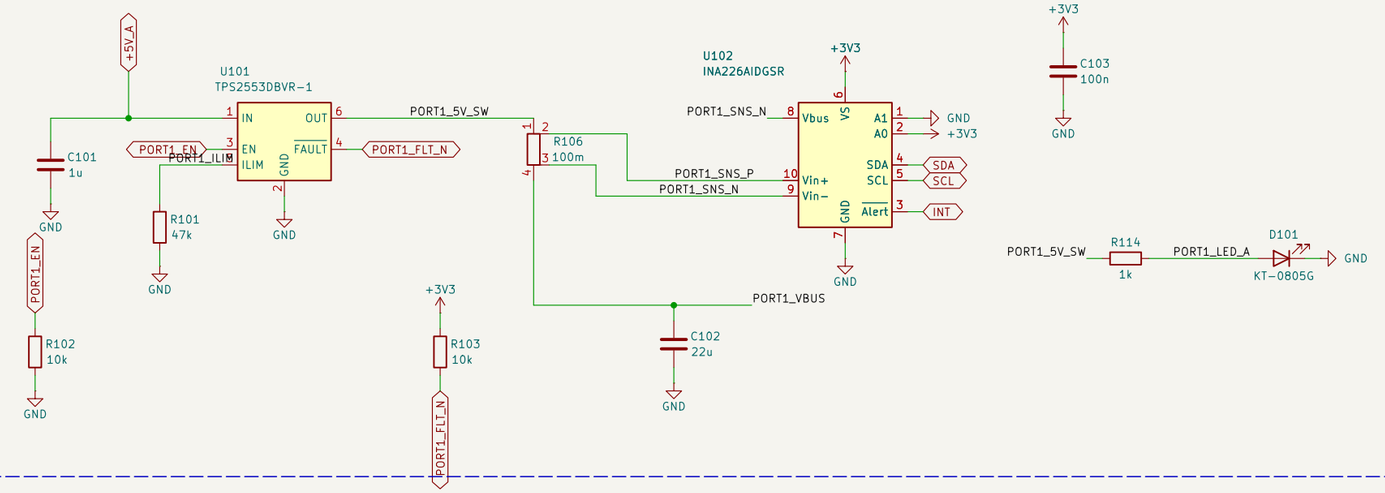
hw-testing-rack

Sheet: /Power/
File: power.kicad_sch

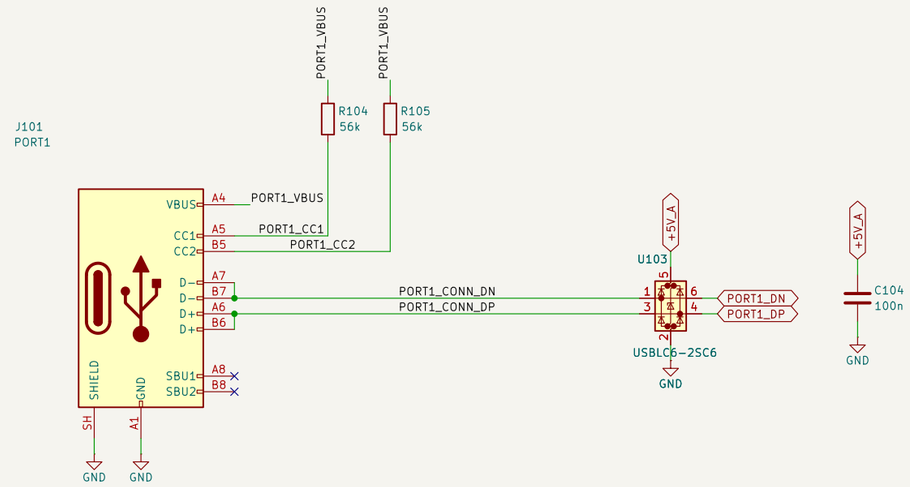
Title: DATA ×6 — 20 V input, two 5 V converters

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 3/9

Port 1 power: +5V_A → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt

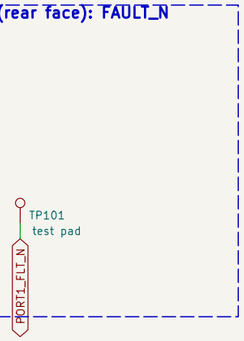


USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 1 = hub port 6, busid 1-1.6; supplied by +5V_A.
INA226 at 0x41; EN and FAULT_N on the TCA9535 U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).

Test point (rear face): FAULT_N



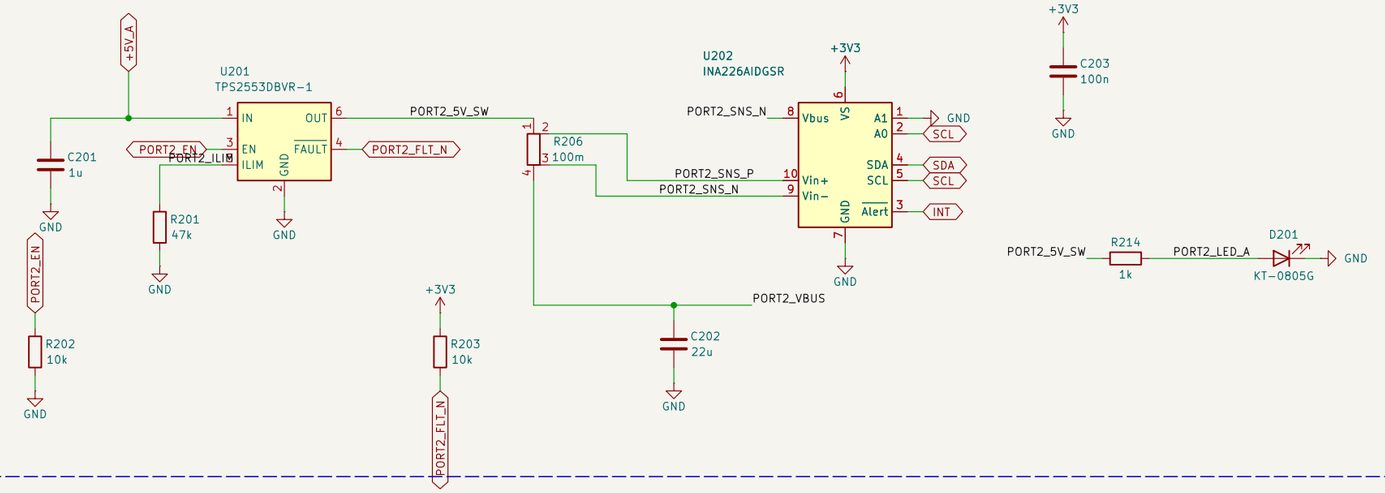
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)
hw-testing-rack

Sheet: /Port 1/
File: port1.kicad_sch

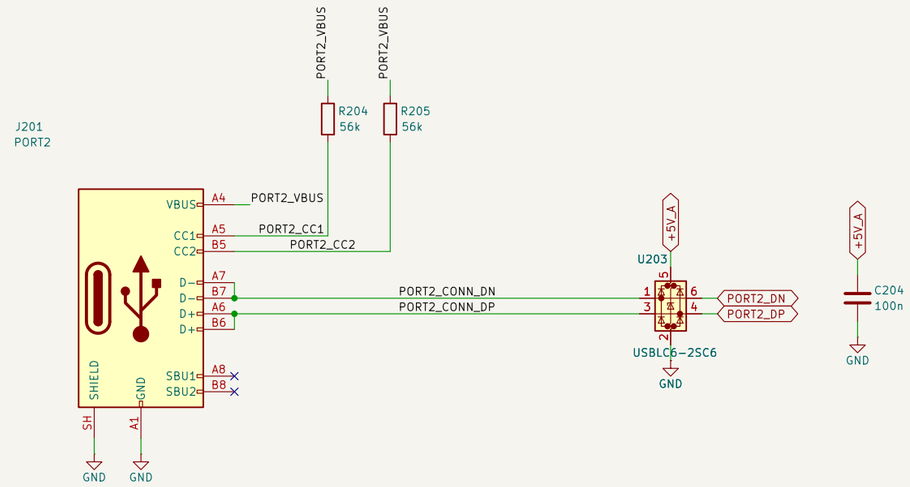
Title: DATA ×6 — USB-C 1: switch, current sense, ESD

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/9

Port 2 power: +5V_A → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt

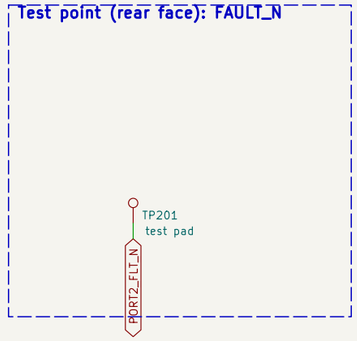


USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 2 = hub port 1, busid 1-1.1; supplied by +5V_A.
INA226 at 0x43; EN and FAULT_N on the TCA9535 U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).

Test point (rear face): FAULT_N



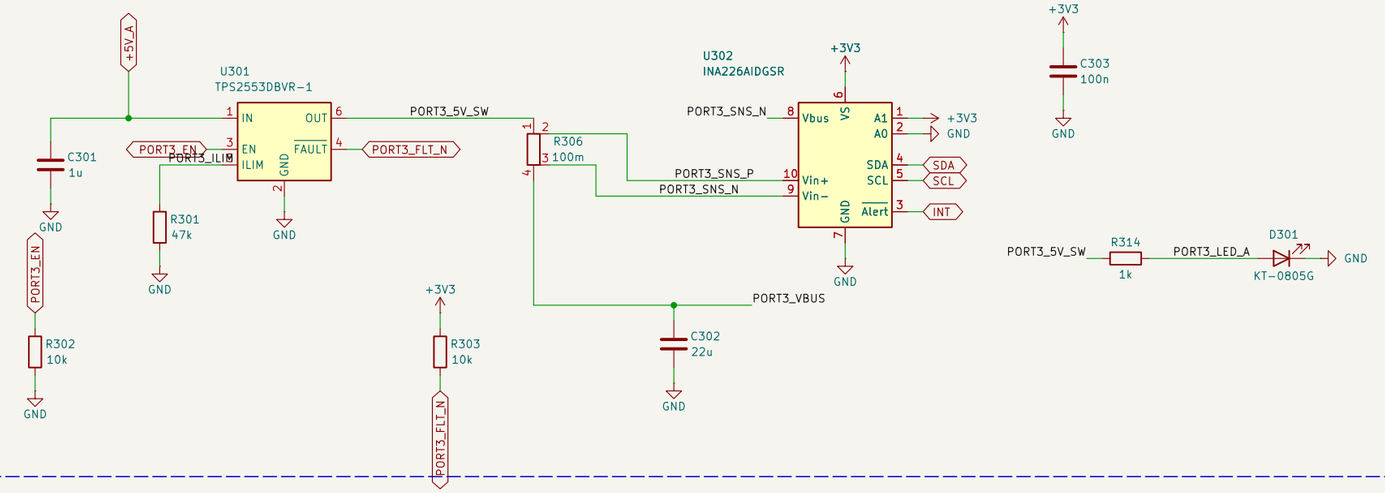
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)
hw-testing-rack

Sheet: /Port 2/
File: port2.kicad_sch

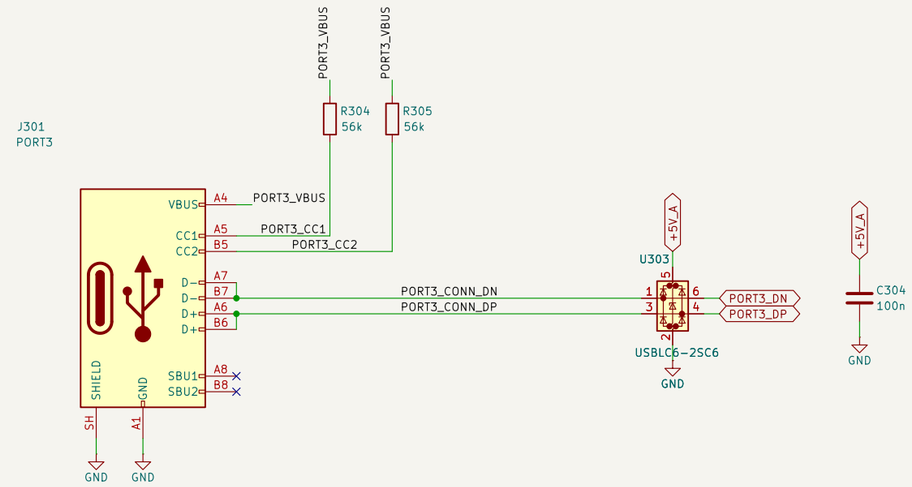
Title: DATA ×6 — USB-C 2: switch, current sense, ESD

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 5/9

Port 3 power: +5V_A → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt

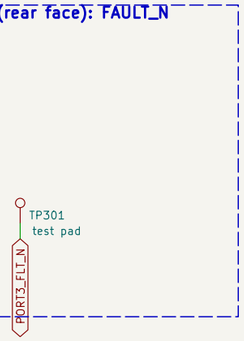


USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 3 = hub port 2, busid 1-1.2; supplied by +5V_A.
INA226 at 0x44; EN and FAULT_N on the TCA9535 U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).

Test point (rear face): FAULT_N



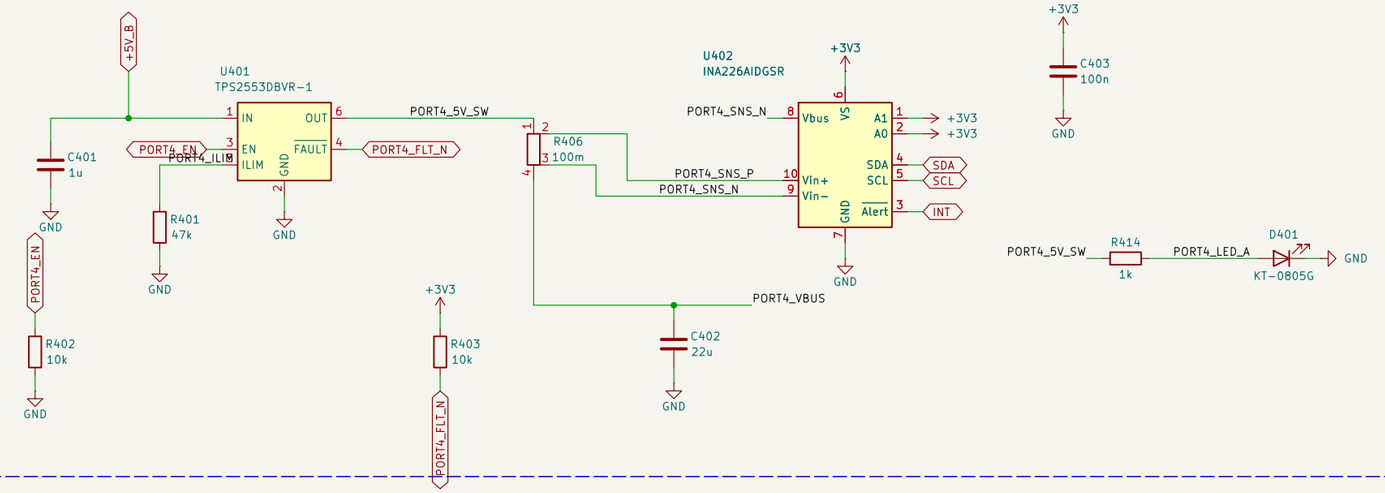
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)
hw-testing-rack

Sheet: /Port 3/
File: port3.kicad_sch

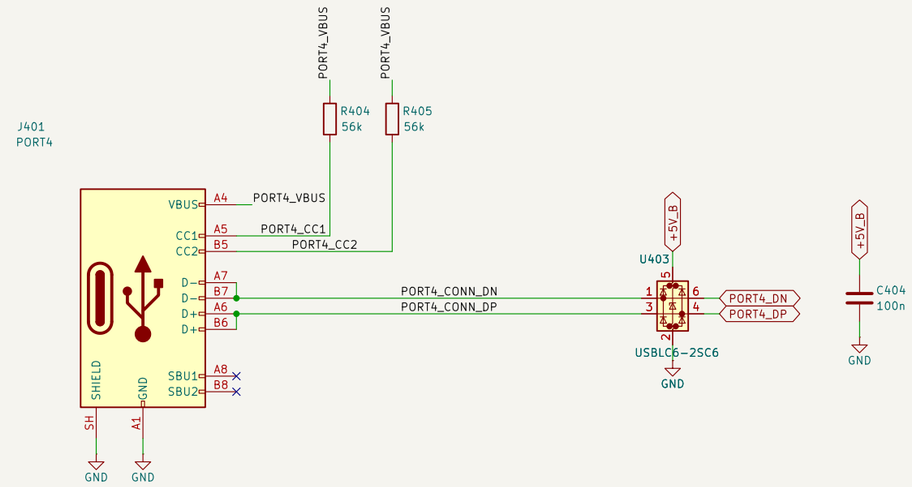
Title: DATA ×6 — USB-C 3: switch, current sense, ESD

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 6/9

Port 4 power: +5V_B → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt

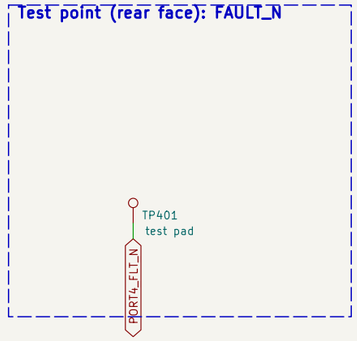


USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 4 = hub port 3, busid 1-1.3; supplied by +5V_B.
INA226 at 0x45; EN and FAULT_N on the TCA9535 U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).

Test point (rear face): FAULT_N



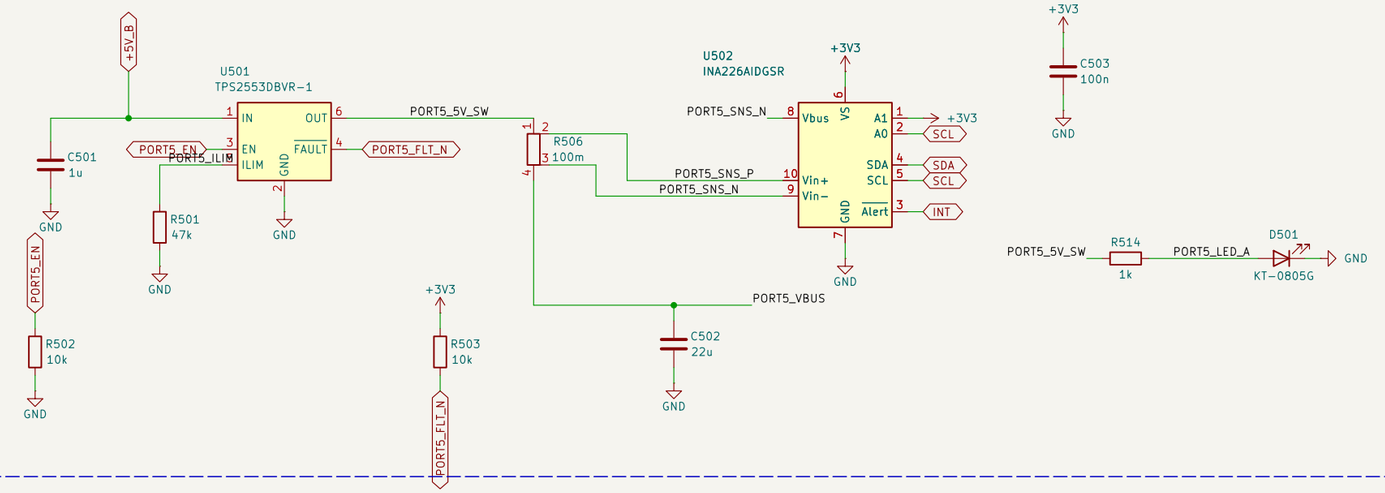
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)
hw-testing-rack

Sheet: /Port 4/
File: port4.kicad_sch

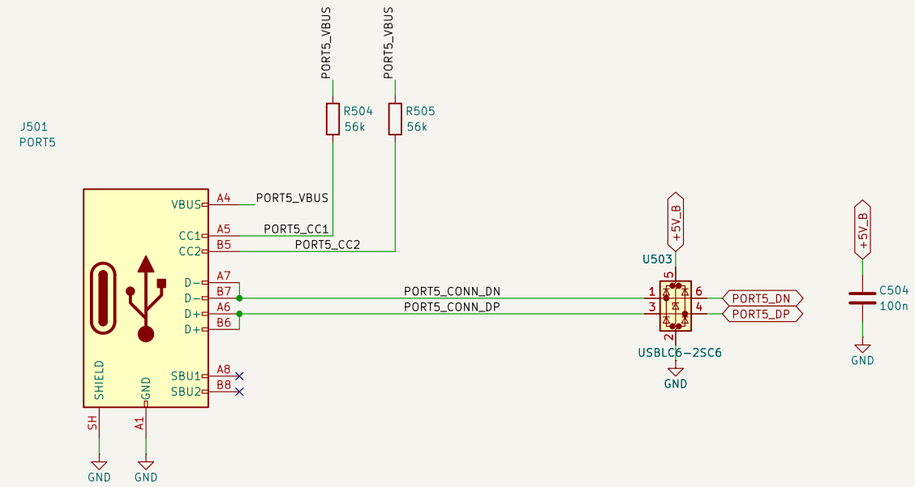
Title: DATA ×6 — USB-C 4: switch, current sense, ESD

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 7/9

Port 5 power: +5V_B → TPS2553-1 → PORT5_5V_SW → 100 mΩ shunt → PORT5_VBUS, the INA226 across the shunt

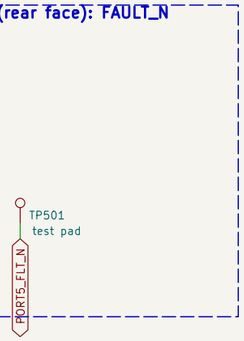


USB-C 5 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 5 = hub port 4, busid 1-1.4; supplied by +5V_B.
INA226 at 0x47; EN and FAULT_N on the TCA9535 U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).

Test point (rear face): FAULT_N



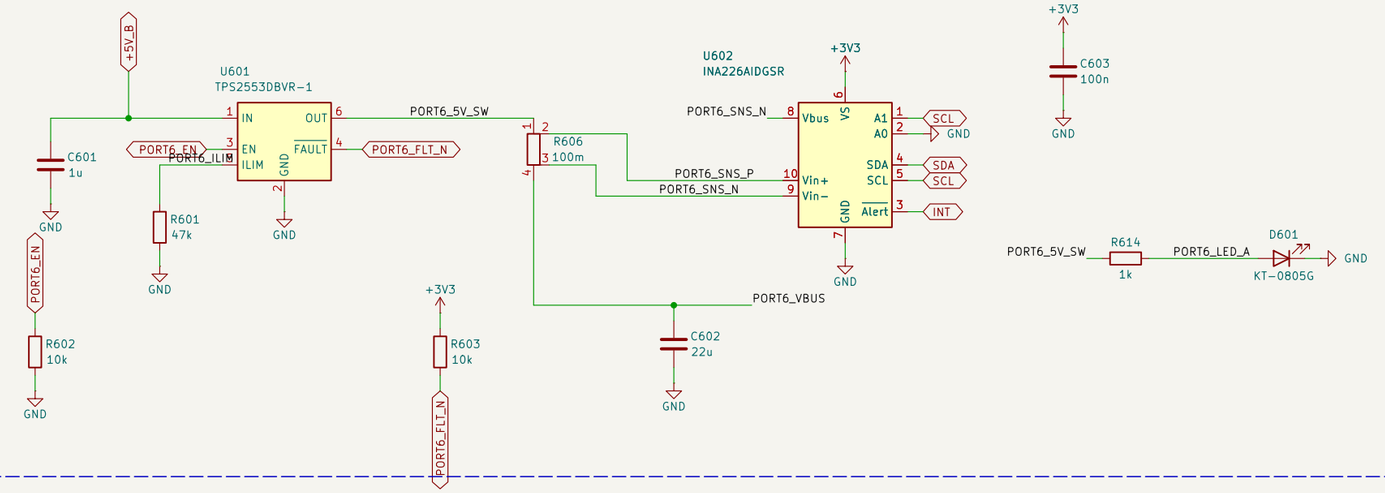
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)
hw-testing-rack

Sheet: /Port 5/
File: port5.kicad_sch

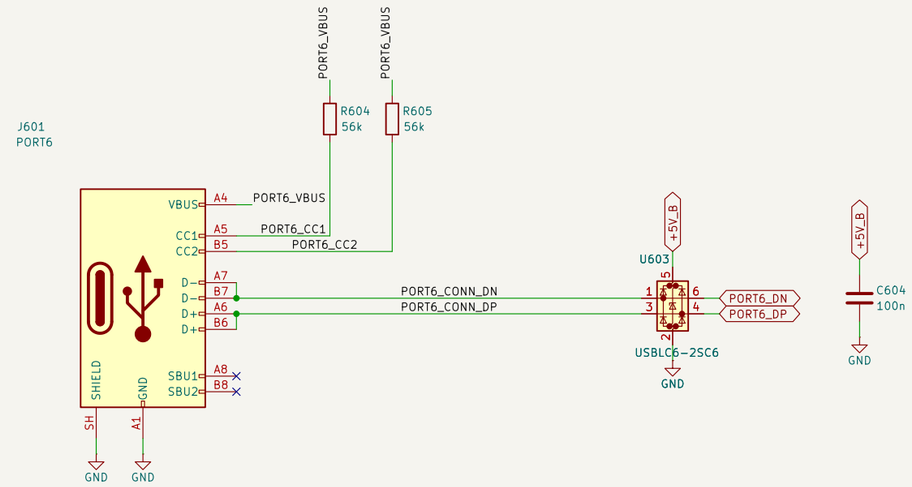
Title: DATA ×6 — USB-C 5: switch, current sense, ESD

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 8/9

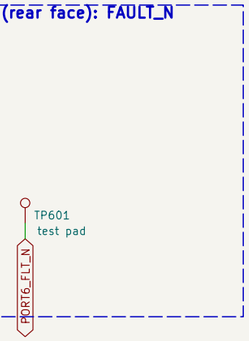
Port 6 power: +5V_B → TPS2553-1 → PORT6_5V_SW → 100 mΩ shunt → PORT6_VBUS, the INA226 across the shunt



USB-C 6 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Test point (rear face): FAULT_N



Port 6 = hub port 5, busid 1-1.5; supplied by +5V_B.
INA226 at 0x4c; EN and FAULT_N on the TCA9535 U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).

Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues drawer board: six switched, measured USB-C behind a hub (hw-testing-rack)

hw-testing-rack

Sheet: /Port 6/
File: port6.kicad_sch

Title: DATA ×6 — USB-C 6: switch, current sense, ESD

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 9/9